DDI-Net 3.0

Explainable drug interaction intelligence

Drug–Drug Interaction Research Report

Metformin + Warfarin

DRUGS ANALYSED

A **Metformin**
DB00331 · approved

B **Warfarin**
DB00682 · approved

Report ID	DDI-20261015-113836-C357C3A4
Generated	15 October 2026, 11:38:36 (UTC, UTC+00:00)
Generated (UTC)	2026-10-15T11:38:36.393Z
Model	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
API report version	3.0
Analysis produced by the service	2026-10-15T11:38:35+00:00
Response received by the browser	2026-10-15T11:38:35.929Z
Service	http://127.0.0.1:8765
Language	English

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RESEARCH USE ONLY

DDI-Net is a research prototype — not a medical device and not clinical decision support. The absence of a documented interaction is not evidence of safety. The BIO-GINE score is a research signal of resemblance to documented pairs, not a probability of clinical harm.

1 Executive summary

No documentation found

No interaction found in the sources checked. This does not mean the combination is safe

The sources checked hold no record. This does not mean the combination is safe.

This is the kind of documentation found, not a level of danger.

IN BRIEF

No interaction record was found in the sources checked (this does not mean the combination is safe). Mechanism hypotheses: 0. BIO-GINE calibrated score: 0.44 — a research signal, not a risk.

FOUR KINDS OF INFORMATION IN THIS REPORT

Documented evidence

none found — this does not mean “safe”

A label or database record that names both drugs.

Mechanism hypotheses

hypotheses: 0; shared Reactome pathways: 1

Assembled from each drug's documented roles separately; not tested for the pair.

Machine-learning prediction

calibrated score 0.44 · regime S2

BIO-GINE's resemblance to documented pairs; not a probability of harm.

Missing or unavailable evidence

FDA labels (openFDA): unavailable

What could not be checked, or is absent from the sources.

What the official sources document

- No interaction record for this pair was found in the sources checked. This does not mean the combination is safe: the sources are incomplete, and some may have been unavailable.

What the quoted evidence directly says

- There is no source text about this pair to quote.

What is only a hypothesis

- The drugs' documented roles form none of the mechanism configurations checked.
- Shared Reactome pathways: 1 (background, not a cause of interaction).

What BIO-GINE adds

- Calibrated score 0.44 (regime S2): the pair moderately resembles pairs documented as interacting. No documentation was found, so the score is the only signal: a lead for research, not evidence that the drugs interact or that they do not. M
- Measured quality on the pooled held-out test (S2 and S3 pairs together; no S2-only estimate exists): AUPRC 0.812 ± 0.010. That is an average over the whole test set, not the reliability of this one prediction.
- One drug of this pair was in the validation set, whose pairs were used to choose the model's settings and to fit the calibration, so the test figures describe this pair only approximately.
- The score says nothing about mechanism, direction, severity or what to do, and it is not a probability of harm (more in section 5).

What remains uncertain

- Source “FDA labels (openFDA)”: unavailable. Its data is not part of this report.

- The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Section 5 gives the quality measured on the held-out test instead — an average over many pairs, not confidence in this pair.
- No source gives a severity, and none is assigned here.
- Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account.

Terms such as S2, AUPRC or CYP450 are explained in the glossary (section 9).

2 Drug information

FIELD	A · METFORMIN	B · WARFARIN
International name (INN)	metformin	warfarin
Russian name	метформин (transliteration)	варфарин (transliteration)
DrugBank ID	DB00331	DB00682
DrugBank groups	approved	approved
Type	small molecule	small molecule
Query as entered	DB00331	DB00682
Matched on	—	—
Protein annotations (targets / enzymes / transporters / carriers)	5 / 0 / 6 / 0	2 / 6 / 0 / 2
Model split (seed 0)	train	val
Protein data available to the model	yes	yes
DrugBank page	https://go.drugbank.com/drugs/DB00331	https://go.drugbank.com/drugs/DB00682

Identifiers: DrugBank. INN: DrugCentral, joined on InChIKey (no value means no match). Russian names: rule-based transliteration — an interface label, not sourced data.

3 Documented interactions

Evidence is grouped by source document: [n] is a document, [n.k] one distinct excerpt of it; the explanation refers to these labels. Several excerpts of one label are one source, not separate confirmations. The English text is the source's own words (an excerpt or DrugBank's standard sentence).

> **No documentation found**
> No interaction record for this pair was found in the sources checked. This does not mean the combination is safe: the sources are incomplete, and some may have been unavailable.

Sources checked

SOURCE	STATUS	DETAIL
DrugBank (TDC export, local)	checked, no match	0 record(s) in the local DrugBank export (TDC, retrieved 2026-08-24).
FDA labels (openFDA)	unavailable	openfda_label: recently unreachable; retrying after a cooldown

4 Biological mechanisms

Below are each drug's documented roles on their own (DrugBank, ChEMBL) and how they line up. These are mechanism hypotheses, not evidence of an interaction: the databases do not record strength of effect or how much of a drug's processing a protein handles.

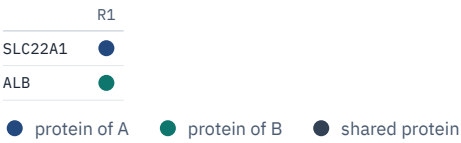
No mechanism data for this pair.

Mechanism hypotheses

The drugs' documented roles form none of the mechanism configurations checked.

Shared biological pathways (Reactome) · 1

1 of 1 shown: pathways through a shared protein first, then the most specific. Protein membership in Reactome pathways (a dot means the protein is in the pathway; colour shows whose protein it is: A, B or shared).



R	PATHWAY	REACTOME	PROTEINS	VIA A	VIA B
R1	Ciprofloxacin ADME	https://reactome.org/content/detail/R-HSA-9793528	5	SLC22A1	ALB

5 BIO-GINE research analysis

CALIBRATED SCORE	RAW SCORE	REGIME	RELIABILITY FOR THIS PAIR
0.44	0.1650	S2	like the held-out pairs of the frozen test

Numbers are rounded; exact values are in Appendix A and the companion JSON.

Model and checkpoint	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
Regime	S2 — one drug was outside the training set
Splits A / B	train / val
Pair's record seen in training	no
Protein data available to the model	A: yes · B: yes

What the two numbers mean

- Raw score (0.1650): the network's direct output, a number between 0 and 1. Near the ends of the scale the values bunch up: differences in the fourth or fifth decimal place mean nothing beyond “very high” (or “very low”).
- Calibrated score (0.44): the same output, rescaled by “temperature scaling”. The rescaling was fitted on validation pairs of which half were documented interactions and half random pairs. It pulls over-confident values back from the edges of the scale but does not change the order of pairs. That is why the two numbers differ.
- Neither number is the probability that the combination harms anyone. The model only learned to tell documented pairs from random ones. The calibrated score could be read as a share of documented pairs only in a half-and-half mix like the one it was fitted on; real drug combinations are not such a mix, and a documented interaction says nothing about how serious it is.
- What they can tell: how strongly the model groups this pair with pairs already documented as interacting. That is useful for ranking pairs for research.

- What they cannot tell: whether an interaction happens in a particular person, how strong or serious it would be, which drug affects which, or what anyone should do.

Measured model quality (held-out test set, not this pair)

METRIC	MEAN ± SD	SEEDS	TEST SET	SOURCE
AUPRC	0.812 ± 0.010	5	pooled drug-disjoint test (S2 and S3; 91% S2)	paper/FACTS.json models.M4_pooled (v2_final_test.csv @ frozen tag)
AUROC	0.818 ± 0.010	5	pooled, both annotated	reports/v3_coverage_reliability.json

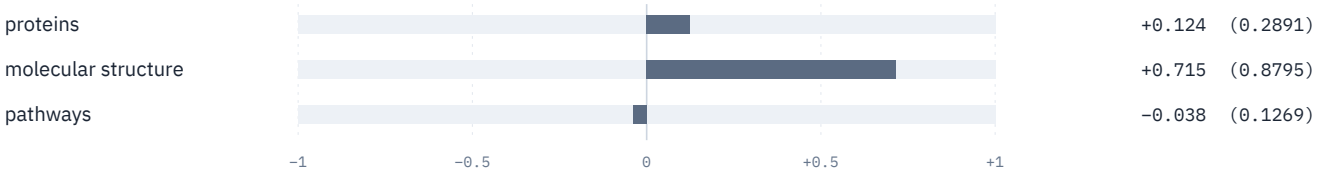
- This pair's regime is S2. The frozen results evaluate S2 pairs only together with S3 pairs: the pooled test has 84,690 pairs, 76,932 of them S2 and 7,758 S3. No S2-only estimate exists, so the pooled figure is shown and labelled as pooled.
- “both annotated”: only test pairs whose two drugs both have protein data (79,712 pairs of the pooled test).
- AUPRC and AUROC describe the model's quality over thousands of test pairs; “±” is the spread between 5 training runs. They do not say how likely this particular prediction is to be right. The service gives no uncertainty for a single pair: it uses one trained copy of the model (seed 0).
- One drug of this pair was in the validation set, whose pairs were used to choose the model's settings and to fit the calibration, so the test figures describe this pair only approximately.

Uncertainty

The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Section 5 gives the quality measured on the held-out test instead — an average over many pairs, not confidence in this pair.

What the model leans on (one kind of data zeroed)

Change in the raw score when one kind of data is removed for both drugs. A minus sign means the score drops without that data. In brackets: the raw score without this data.



Raw score: 0.1650

This shows which data the model leans on for this pair, not why the drugs interact: the model never saw zeroed inputs in training.

Method check: reproduces the frozen V2 analysis on 84,690 test rows, max |Δ| = 6.1e-6.

Method: branch zeroed before the frozen fusion layer for both drugs; mask unchanged (as scripts/49_v2_interpretability.py at the frozen tag)

Notes from the service

- One drug was in the validation set: the model was not trained on it, but it was used to choose the configuration and the calibration.
- The calibrated score was fitted on validation pairs with equal numbers of documented and randomly drawn pairs. It is not a probability of harm, nor how often interactions happen in practice.

6 Structured explanation from the service

The text below was written by the service with deterministic templates over the retrieved records (no language model) and checked by it for advice wording. Labels point to records: [n] source document, [n.k] one of its excerpts, H hypothesis, M model, R Reactome pathway. The text below comes from the service; its terms (S2, AUPRC, AUROC) are explained in the glossary (section 9).

What is known

Missing — No record of an interaction between metformin and warfarin was found in the sources checked (DrugBank (TDC export)). This does not mean the combination is safe: for many pairs this data was never collected, and labels often describe interactions for whole drug classes.

Why this may happen

Missing — The documented roles of these drugs share no protein through which they might affect each other. The mechanism may be different, or not yet studied.

Annotation — Proteins of both drugs take part in 1 shared biological process(es) in Reactome — for example “Ciprofloxacin ADME”. A shared process is background, not a cause of interaction. R

What the AI found

Model — BIO-GINE finds this pair moderately resembles pairs already recorded as interacting. This is resemblance to recorded pairs, not an estimate of harm. M

Model — For this pair the model's score depends most on the molecular structure: without it the score moves by 0.71. This says what the model leans on, not why the drugs interact. M

Caveat — One drug was in the validation set: the model was not trained on it, but it was used to choose the configuration and the calibration. M

Caveat — The calibrated score was fitted on validation pairs with equal numbers of documented and randomly drawn pairs. It is not a probability of harm, nor how often interactions happen in practice. M

How certain is this?

Caveat — No severity is assigned: the DrugBank export has none, and for labels only the section that names the drug is shown.

Model — Model quality on the pooled held-out test (S2 and S3 pairs together, 91% S2; the frozen results hold no S2-only figure): AUPRC 0.812 ± 0.010 (spread across 5 training runs). That is an average over many pairs, not confidence in this particular pair. M

What is missing

Missing — The source “FDA labels (openFDA)” could not be checked right now. Its data is not part of this answer.

Caveat — Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account — everything that decides whether an interaction shows up and how strongly.

Caveat — Decisions about medicines belong to a doctor or pharmacist. This page cannot replace them.

Method: deterministic templates over retrieved records; no language model

7 Sources and reproducibility

Evidence records

No documentation found.

Provenance and parameters

Endpoint (report)	`http://127.0.0.1:8765/api/v3/report`
Request body (report)	`{"drug_a":"DB00331","drug_b":"DB00682","include_labels":true}`
Response received by the browser (report)	2026-10-15T11:38:35.929Z
Report generated by the service (generated_at)	2026-10-15T11:38:35+00:00
API report version	3.0
Explanation method	deterministic templates over retrieved records; no language model
Model	BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0
Frozen research tag	`v2-final-github-safe-2026-09-03 · 92c481eeaba8faff991ced850e1c4de418ea31b0`
Site	http://127.0.0.1:4173
Site build commit	not recorded
PDF generator	DDI-Net web, in-browser export (@react-pdf/renderer)
Generated	15 October 2026, 11:38:36 (UTC, UTC+00:00)
Generated (UTC)	2026-10-15T11:38:36.393Z
Language	English
SHA-256 of the response data	`c357c3a4b2ee980a0c1204549a60d757b85b27732725e3e5a8bdff1f172849e4` Hash of the service response(s) as canonical JSON (keys sorted, serialised with JSON.stringify). The same hash is in the companion JSON file, available from the “JSON” button next to “Export PDF”.

Retrieval times are given per record (retrieved_at) as returned by the service: the export date for local extracts, the request date for labels.

8 Limitations

Core limitations

- DDI-Net is a research tool, not clinical decision support and not a medical device. Decisions about medicines belong to a doctor or pharmacist.
- Unlabelled pairs are not confirmed safe. A missing record means only that the sources checked do not contain one.
- BIO-GINE scores are not calibrated probabilities of clinical harm: temperature scaling was fitted on validation pairs at 50% prevalence against randomly drawn unlabelled pairs.
- The drug order in the DrugBank (TDC) export was verified by measurement for only two interaction types; for the others it is either meaningless (symmetric types) or unverified, so directional claims (“A raises the level of B”) are not made.
- External medical evidence (FDA labels, RxNorm) may be incomplete or unavailable at query time; its status is listed under “Sources checked”.

Limitations reported by the service for this report

- Only the 1,705 drugs of the research dataset; 1,497 of them have an international name.
- DrugBank records carry no severity and do not state their basis. For many record types the drug order in the export is unreliable (measured).
- Labels are searched by drug name. Statements about whole drug classes (“strong CYP3A4 inhibitors”) are not recognised, so no match does not mean the label is silent.

- Mechanism hypotheses are assembled from each drug's roles separately; strength of effect and the protein's share of the work are unknown.
- The BIO-GINE score is a research signal of resemblance to documented pairs. It is not a probability of harm.

> RESEARCH USE ONLY
> This is a research prototype, not a medical device. It does not replace a doctor or pharmacist and cannot say whether a combination is safe for a particular person.

9 Glossary

Plain explanations of the terms used in this report.

Source evidence vs model prediction	Source evidence is text that an official label or a database actually contains about these two drugs. A model prediction is a number BIO-GINE computes from patterns in the data it was trained on. The first records what is known; the second is a research guess.
FDA label (SPL)	A medicine's official prescribing information in the United States, published by the FDA as Structured Product Labeling. One label is one document with its own set ID, divided into sections such as Warnings or Drug Interactions.
Document and excerpt	An excerpt is one sentence quoted from a document. Several excerpts of one label are still one source and do not confirm each other. When a label repeats a sentence in two sections, it is shown once.
DrugBank (TDC export)	A large database of drugs and known interactions; DDI-Net uses the public copy distributed by Therapeutics Data Commons. A record states only an interaction type, in a standard sentence, with no severity and no note of what it is based on.
Direction	Which drug affects which. For most interaction types the drug order in the DrugBank export is unreliable, so the report names no direction from DrugBank. Label sentences are quoted as written; no direction is inferred from the mere fact that one drug's label names the other.
Cytochrome P450 (CYP450)	A family of enzymes, mostly in the liver (for example CYP2C9, CYP3A4), that break down many medicines. If one drug slows an enzyme that breaks down another, the second may stay in the body longer.
Substrate, inhibitor, inducer	A substrate is a drug a protein processes. An inhibitor slows the protein down. An inducer makes the body produce more of it.
Mechanism hypothesis	A possible biological explanation assembled from what databases record about each drug separately (for example: drug X inhibits an enzyme that processes drug Y). It was not tested for this pair and says nothing about the size of any effect.
Reactome pathway	A curated chain of reactions in the cell. Two drugs sharing a pathway is background information, not a cause of interaction.
Raw and calibrated score	The raw score is the model's direct output between 0 and 1; the calibrated score is the same output rescaled on balanced validation pairs. Both put pairs in the same order. Neither is a probability of harm.
Regimes S1, S2, S3	Whether the model saw the drugs during training. S1: both drugs were in training (quality not measured). S2: one drug was not in training. S3: neither was — the hardest test.
Held-out test; "pooled"	Pairs set aside in advance and never used for training; quality is measured on them. "Pooled" means S2 and S3 test pairs measured together (84,690 pairs, 7,758 of them S3).
AUROC	The chance that the model ranks a randomly chosen documented pair above a randomly chosen pair from the comparison set (random pairs without a record). 0.5 is guessing; 1 is a perfect ranking.
AUPRC	Area under the precision–recall curve: how well the model finds documented pairs without raising many false alarms, over all thresholds at once. Half of the test pairs are documented, so guessing gives about 0.5; 1 is perfect.
± (spread between runs)	The model was trained 5 times, each from a different random starting point (seed). "±" is the standard deviation of the result between those runs. It is not an error bar for a single pair.

A Appendix A. Machine-readable summary (JSON)

Key fields as JSON for automated checking. The full service responses are in the companion JSON file (same SHA-256).


```
{
  "schema": "ddinet.pdf-summary/1",
  "report_id": "DDI-20261015-113836-C357C3A4",
  "generated_at": "2026-10-15T11:38:36+00:00",
  "generated_at_utc": "2026-10-15T11:38:36.393Z",
  "time_zone": "UTC",
  "language": "en",
  "payload_sha256": "c357c3a4b2ee980a0c1204549a60d757b85b27732725e3e5a8bdff1f172849e4",
  "inputs": [
    {"endpoint":"http://127.0.0.1:8765/api/v3/report","request":{"drug_a":"DB00331","drug_b":"DB00682","include_labels":true},
      "received_at":"2026-10-15T11:38:35.929Z"}
  ],
  "server_generated_at": "2026-10-15T11:38:35+00:00",
  "report_version": "3.0",
  "drugs": [
    {"role":"A","id":"DB00331","inn":"metformin","name_ru":"метформин"},
    {"role":"B","id":"DB00682","inn":"warfarin","name_ru":"варфарин"}
  ],
  "evidence_status": "no_documented_evidence_found",
  "evidence_counts": {
    "labelDocuments": 0,
    "labelExcerpts": 0,
    "noEffectExcerpts": 0,
    "labelRecords": 0,
    "mergedRepeats": 0,
    "databaseRecords": 0
  },
  "source_documents": [],
  "evidence": [],
  "conflicts": 0,
  "hypotheses": [],
  "shared_pathways_total": 1,
  "model": {
    "model": "BIO-GINE M4",
    "checkpoint": "bd45f84e3c1b2c33",
    "available": true,
    "calibrated_score": 0.44394229455360606,
    "raw_score": 0.16501888092486197,
    "regime": "S2",
    "reliability": "held_out_like",
    "abstain": false
  },
  "sources_checked": [
    {"source":"drugbank_tdc","status":"no_match"},
    {"source":"openfda_label","status":"unavailable"}
  ]
}
```